Investigation of Delayed Neutron Precursor Data Uncertainties for Microscopic and Summation Calculations

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Abstract

This paper presents an analysis on the sensitivity of delayed neutron precursor (DNP) group parameters, group half-lives τ_k and yields $\nu_{d,k}$, and summed delayed neutron yields to uncertainties in their nuclear data for various datasets, including the International Atomic Energy Agency (IAEA) beta-delayed neutron emission database, ENDF/B-VII.1, ENDF/B-VIII.0, JEFF-3.1.1, and JENDL-5. These uncertainty-weighted sensitivities provide a detailed view of which DNPs need more accurate data such that analyses using individual DNPs, such as microscopic group parameter formulation and summation calculations, may be more accurate. Improving the accuracy of these analyses is of particular importance for kinetics models, which rely on accurate DNP group parameters. The microscopic approach for group parameter calculation is useful for providing insight into how each DNP contributes to delayed neutron emission, but it requires accurate nuclear data. The analyses conducted in this work reveal several DNPs that correlate strongly with the DNP group parameters, indicating a need for more accurate data for those DNPs. There are also DNPs which have large uncertainties and vary significantly between datasets which also warrant additional attention. This paper informs which DNPs are in need of updates to their data for various nuclear datasets.

Keywords — Delayed Neutron Precursors, Group Parameters, Nuclear Data, Nuclear Data Uncertainties, ENDF, JEFF

I. INTRODUCTION

Modeling and simulation of nuclear reactors enables more accurate predictions about the reactor behavior during transient events, such as removal of a control rod. To accurately model these dynamics, several different effects must be included: the change in the neutron density, the various feedback effects that influence the reactivity, the emission of prompt neutrons, and the emission of delayed neutrons. These effects are well understood and have been simplified using various approximations. For example, the delayed neutrons are treated as having six to eight sources, or DNP groups, even though in reality there are hundreds of precursor nuclides that decay and emit these delayed neutrons [1].

There are several different methods for calculating these DNP groups, and in this work the microscopic approach is of particular interest [2, 3, 4, 5, 6, 7, 8, 9]. This approach uses experimental data for individual DNPs combined together computationally. One strength of this approach is that it provides insight into how each individual DNP contributes to the DNP group parameters. However, this strength is also a weakness, as it relies on the nuclear data for every DNP individually, which means a few large uncertainties in the various DNPs can lead to uncertain results for the DNP group parameters. There is also a related, and similar approach, called the summation approach. This approach uses the same data, and is used for calculation of summed parameters such as the total delayed neutron yield ν_d and the average half-life $\bar{\tau}$.

There have been investigations in the literature that use the microscopic and summation approaches to evaluate sensitivities and compare different datasets [10, 9]. Although these analyses are informative and show many of the differences between datasets, they do not discuss the importance of DNPs in regards to their effect on the calculated DNP group parameters but instead based on their contribution to the total delayed neutron yield. Generally the larger the delayed neutron yield of a DNP, the more important it is, but these analyses do not investigate how the uncertainty in each DNP propagates through to calculated DNP group parameters.

This work seeks to expand upon these previous analyses by using the microscopic approach in combination with Monte Carlo stochastic sampling of nuclear data uncertainties to evaluate uncertainty-weighted sensitivities of individual DNP's to DNP group half-lives and delayed neutron yields. These results indicate which DNPs in different datasets have large impacts on the DNP group parameters, as well as which DNP have both large impacts and large uncertainties relative to those impacts. By providing the data of importance for these datasets, future analyses can target experiments to produce those specific data to further improve DNP group parameters and kinetics models.

II. METHODS

This section details the three stages to the analyses presented in this paper: the data treated as an input, the solver which calculates the data of interest, and the data analysis which details how the solver results are processed. The solver itself can be considered a converter of the DNP nuclear data into six group DNP parameters, which are then passed into data analysis calculations.

II.A. Data

There are two different sets of data used in this work: the nuclear data for each individual DNP and the calculated DNP group parameters that approximate those individual DNPs. The I DNP have the following data:

$$\begin{aligned} CFY_i &= \text{the cumulative fission yield of DNP } i \text{ [-]} \\ P_{n,i} &= \text{the emission probability of DNP } i \text{ [-]} \\ \tau_i &= \text{the half-life of DNP } i \ [s] \\ \chi_{d,i} &= \text{the delayed neutron energy spectrum from DNP } i \text{ [-].} \end{aligned}$$

The cumulative fission yield represents the total production of that nuclide for a single fission, including decays that eventually lead to that nuclide. The emission probability is the probability for a delayed neutron to be emitted from the DNP[a]. For this work, the delayed neutron energy spectra are not investigated, as it is considered out of scope.

The DNP group parameters have the following data:

$$\begin{aligned} \nu_{d,k} &= \text{the delayed neutron yield of group } k \text{ [-]} \\ \tau_k &= \text{the half-life of group } k \ [s] \\ \chi_{d,k} &= \text{the delayed neutron energy spectrum from group } k \text{ [-].} \end{aligned}$$

[a]This can alternatively be considered the number of delayed neutrons emitted per radioactive decay.

The delayed neutron yield for each group $\nu_{d,k}$ combines the emission probabilities and cumulative fission yields, representing the number of delayed neutrons emitted from group k from a single fission.

The datasets used in this work are given in Table I. The IAEA beta-delayed neutron emission database does not have cumulative fission yield data combined with it, though analyses using the data use JEFF-3.1.1 cumulative fission yields [11]. For the analyses in this work, the cumulative fissions yields from all of the datasets will be used and compared against to offer a comprehensive analysis.

TABLE I
Various data sources as well as the data which they contain.

			Available Data		
Data name	Year	Source	τ_i	$P_{n,i}$	CFY_i
IAEA database	2018	[11]	Yes	Yes	No
Joint Evaluated Fission and Fusion (JEFF) 3.1.1	2017	[12]	Yes	Yes	Yes
Evaluated Nuclear Data File (ENDF)/B-VII.1	2011	[13]	Yes	Yes	Yes
ENDF/B-VIII.0	2018	[14]	Yes	Yes	Yes
Japanese Evaluated Nuclear Data Library (JENDL)-5	2021	[15]	Yes	Yes	Yes

The data of interest in these datasets are for the individual DNP nuclear data. This data is passed into the solver, which then converts these data into DNP group parameters for analysis and post-processing.

II.B. Solver

For the microscopic and summation methods, there are two distinct solution methods. The summation method is the simpler of the two, and simply involves summing the nuclear data to calculate the total delayed neutron yield and average half-life. The microscopic method is more in-depth, as it is used to calculate the DNP group parameters, which give time-dependency and can be used to calculate the total delayed neutron yield and average half-life.

Equations (1) and (2) are the equations for the total delayed neutron yield and average half-life, respectively, calculated using the summation approach as:

$$\nu_d = \sum_{i=1}^{I} P_{n,i} CFY_i, \tag{1}$$

and

$$\bar{\tau} = \frac{1}{\nu_d} \sum_{i=1}^{I} CFY_i P_{n,i} \tau_i. \tag{2}$$

The uncertainties for each calculation can be propagated as shown in Equations (3) and (4) as:

$$\Delta\nu_d^2 = \sum_{i=1}^{I} (\Delta P_{n,i} CFY_i)^2 + (P_{n,i} \Delta CFY_i)^2, \tag{3}$$

and

$$\begin{aligned} \Delta\bar{\tau}^2 = \frac{1}{\nu_d^2} \sum_{i=1}^{I} & (\Delta CFY_i P_{n,i} \tau_i)^2 + (CFY_i \Delta P_{n,i} \tau_i)^2 \\ & + (CFY_i P_{n,i} \Delta\tau_i)^2 + \frac{1}{\nu_d^2} (CFY_i P_{n,i} \tau_i \Delta\nu_d)^2. \end{aligned} \tag{4}$$

These calculations are automated in the solver using the uncertainties Python package [16]. The standard deviation propagated in the summation calculation are given by the dataset used, and any values without a standard deviation or a standard deviation of 0 use a value of 10^{-12}.

The microscopic approach enables calculation of time-dependent parameters by calculating the delayed neutron count rate from each DNP after irradiation of a fissile nuclide. The solver first calculates the concentration of each DNP, followed by the delayed neutron count rate, and finally uses the Scipy optimize module to calculate the six DNP group parameters that optimally fit that delayed neutron count rate [17]. To calculate sensitivities and propagate uncertainties, a Monte Carlo sampling technique is used [9]. The solution is calculated with tolerances of 10^{-11} for ten randomly sampled initial conditions for the nominal nuclear data, which is then passed as an initial guess for each of the sampled Monte Carlo solves. The concentrations have the uncertainty propagated using the uncertainties package, and then in the count rate calculation the concentration N_i, half-life τ_i, and emission probability $P_{n,i}$ are sampled using a normal distribution with a different random value for each DNP.

The concentration of each DNP is calculated assuming the sample has reached equilibrium[b]

[b]The parasitic absorption and transmutation cross-sections of the DNPs are negligible and are not included in

[8]:

$$N_i(t=\infty) = \frac{CFY_i}{\lambda_i}, \tag{5}$$

where

$$\lambda_i = \frac{\ln 2}{\tau_i}$$
$$= \text{the decay constant for DNP } i \ [s^{-1}].$$

The delayed neutron count rate can then be calculated as shown in Equation (6) as:

$$\dot{n}_d(t) = \sum_{i=1}^{I} P_{n,i} N_i(t=\infty) e^{-\lambda_i t}. \tag{6}$$

The count rate is first calculated using the nominal nuclear data before this calculation is repeated using the independently, normally sampled nuclear data for each DNP.

Once the count rate is calculated, a non-linear least squares calculation can be used to determine the combination of DNP group parameters most optimally fits the calculated delayed neutron count rate. The coefficient matrix and solution vector for the non-linear least squares solve are based on the DNP group parameter fit for the delayed neutron count rate, given in Equation (7) as [5]:

$$\dot{n}_d(t) = \sum_{k=1}^{K} \nu_{d,k} e^{-\lambda_i t}. \tag{7}$$

Once the delayed neutron count rate is acquired, an over-determined non-linear least squares approach is used to calculate the group yields and decays [1, 18, 19]. Equation (8) gives the general form of the least squares equation as:

$$A\vec{x} \approx \vec{b}, \tag{8}$$

this equation.

where

$$A = \text{the coefficient matrix [-]}$$
$$x = \text{the solution vector } [\# \cdot s^{-1}]$$
$$b = \text{the target vector } [\# \cdot s^{-1}].$$

Equation (9) gives the coefficient matrix as:

$$A = \begin{bmatrix} e^{-\lambda_1 t_0} & e^{-\lambda_2 t_0} & \dots & e^{-\lambda_K t_0} \\ e^{-\lambda_1 t_1} & e^{-\lambda_2 t_1} & \dots & e^{-\lambda_K t_1} \\ \vdots & \vdots & \ddots & \vdots \\ e^{-\lambda_1 t_m} & e^{-\lambda_2 t_m} & \dots & e^{-\lambda_K t_m} \end{bmatrix}. \tag{9}$$

The solution vector contains only the group yields, given in Equation (10) as:

$$\vec{x} = \begin{bmatrix} \bar{\nu}_{d,1} \\ \bar{\nu}_{d,2} \\ \vdots \\ \bar{\nu}_{d,K} \end{bmatrix}. \tag{10}$$

The target vector contains the delayed neutron count rate data for m time-steps, shown in Equation (11) as:

$$\vec{b} = \begin{bmatrix} \dot{n}_d(t_0) \\ \dot{n}_d(t_1) \\ \vdots \\ \dot{n}_d(t_m) \end{bmatrix}. \tag{11}$$

Using least squares with a non-relative residual would lead to early time-step count rates dominating due to their larger magnitudes. This is resolved by minimizing a relative residual, given by Equation (12) as:

$$r = \min_x \left|\left| \frac{\vec{b} - A\vec{x}}{\vec{b}} \right|\right|_2, \tag{12}$$

where the Euclidean norm, $||\cdot||_2$, is defined in Equation 13 as:

$$||\vec{x}||_2 = \sqrt{x_1^2 + \cdots + x_n^2}. \tag{13}$$

The relative residual is used instead of a chi-squared statistic because the Monte Carlo sampling method is already in place to calculate and propagate the uncertainties [20].

The DNP group parameters can then be used to evaluate the total delayed neutron yield and average half-life, similar to Equations (1) and (2). This is shown in Equations (14) and (15) as:

$$\nu_d \approx \sum_{k=1}^{K} \nu_{d,k}, \tag{14}$$

and

$$\bar{\tau} \approx \frac{1}{\nu_d} \sum_{k=1}^{K} \nu_{d,k}\tau_k. \tag{15}$$

These equations are approximate because the number of groups K used is fewer than the total number of DNPs I. To differentiate between the calculation of these parameters using I DNPs and K groups, the parameters will be written as $\nu_d(I)$ and $\bar{\tau}(I)$ when calculated using I DNPs and $\nu_d(K)$ and $\bar{\tau}(K)$ when calculated using K groups.

II.C. Data Analysis

The Pearson correlation coefficient (PCC) is used to analyze the DNP group parameters calculated from the solver for various unique combinations of sampled DNP nuclear data and is calculated using the SciPy stats module [17]. The PCC measures the linear correlation between the DNP data and the calculated DNP group parameters. A measure of 0 indicates there is no correlation, a measure of +1 indicates perfect positive linear correlation, and a measure of -1 indicates perfect negative linear correlation. So, for example, ^{87}Br can be expected to have a PCC near +1 for the impact of its half-life τ_i on the half-life of group one[c] τ_1. This is because ^{87}Br is the longest-lived DNP[d], so the longest-lived DNP group will be highly dependent on its nuclear data.

[c] Group one is the longest-lived DNP group.

[d] There are other DNPs with longer half-lives, such as ^{211}Tl, ^{210}Tl, and ^{210}Hg, but they live only tens of seconds longer and generally have negligible concentrations.

The PCC was selected to measure variation in the group parameters because the sensitivities did not generally have strong non-linear behavior. The non-linear least squares will result in non-linear effects, but linear correlations dominate in the observed sensitivities. The PCCs are calculated for each DNP group half-life τ_k and delayed neutron yield $\nu_{d,k}$ based on changes in each DNPs' concentration N_i, half-life τ_i, and emission probability $P_{n,i}$. This can be represented as $PCC_{i,v,k}$, where i represents the DNP i; v represents the DNP value, such as the half-life, emission probability, or concentration; and k represents the DNP group.

To improve this analysis, some of this data was combined to form more concise coefficients. The total PCC for a specific DNP i is given in Equation (16) is represented as:

$$PCC_i = \sum_v \sum_k |PCC_{i,v,k}| \,, \tag{16}$$

where

v = the DNP value, representing the half-life, emission probability, or concentration of DNP i [-]

$PCC_{i,v,g}$ = the PCC for DNP i and value v with DNP group parameter k [-].

These PCCs can be combined with the relative uncertainty associated with each DNP to capture the nuclides which both impact the DNP group parameters most significantly and have large uncertainties. Specifically, this is calculated in Equation (17) as the relative uncertainties times the magnitude of the PCC for a DNP i and each DNP value v as U_i:

$$U_i = \sum_v \frac{\sigma_{i,v}}{x_{i,v}} \sum_k |PCC_{i,v,k}| \,, \tag{17}$$

where

$U_i =$ the total uncertainty-weighted sensitivity from the contribution of every DNP i [-]

$\sigma_{i,v} =$ the uncertainty in the DNP value v for DNP i [-]

$x_{i,v} =$ the DNP value v, representing the half-life, emission probability, or concentration of DNP i [-].

Equation (17) shows that the relative uncertainty for each nuclear data component that is randomly sampled, such as the emission probability, is multiplied by the summed magnitudes of the PCC for each DNP group parameter using that data. These values are then all summed, giving a total measure of the relative uncertainty-weighted PCC for each DNP, which indicates how much attention that specific DNP needs. This metric is larger when:

- changes in the DNP's nuclear data cause large changes in the DNP group parameter(s)
- the relative uncertainty in the DNP's nuclear data is large.

This can also be represented for any individual DNP value by removing the summation over v. This is shown in Equation (18) which gives the uncertainty-weighted PCC for a given DNP i and nuclear data v as:

$$U_{i,v} = \frac{\sigma_{i,v}}{x_{i,v}} \sum_k |PCC_{i,v,k}| \,. \tag{18}$$

III. RESULTS

This section presents the results for different datasets of cumulative fission yields, emission probabilities, and half-lives. Each sub-section will contain the following analyses for each dataset:

- the total delayed neutron yield calculated from summation of DNPs $\nu_d(I)$ and from the group parameters $\nu_d(K)$
- the average half-life calculated from summation of DNPs $\bar{\tau}(I)$ and from the group parameters $\bar{\tau}(K)$
- the six DNPs with the largest yield contributions

TABLE II
Default parameters for each microscopic calculation.

Parameter	Value	Units
Fissile Nuclide	^{235}U	[-]
Incident Neutron Energy	0.0253	$[eV]$
Final Decay Time	600	$[s]$
Decay Time Spacing	Logarithmic	[-]
Decay Time Nodes	800	[-]
Sampled Simulations	5000	[-]

- the tabulated ten largest PCCs and uncertainty-weighted PCCs
- a chart of nuclides made using The Advanced Reactor Modeling Interface (ARMI) with summed PCCs and uncertainty-weighted PCCs [21]
- a bar chart with the most significant uncertainty-weighted PCCs.

Table II shows the default parameters used for each simulation.

As shown in Table I, the IAEA database does not include cumulative fission yield data. For that analysis, the JENDL-5 cumulative fission yields are used since they are the most recent data. However, part of the analysis will also investigate the effects of hybrid datasets that vary the emission probability, half-life, and cumulative fission yield data.

This work is also verified with total delayed neutron yields from a related IAEA Coordinated Research Project (CRP) analysis [11]. They present a benchmark problem using JEFF-3.1.1 cumulative fission yields with ENDF/B-VIII.0 emission probabilities and half-lives, which is replicated in this work [22, 14]. The four groups in the study are The French Alternative Energies and Atomic Energy Commission (CEA), The Centre for Energy, Environmental and Technological Research (CIEMAT), the Japan Atomic Energy Agency (JAEA), and the University of Nantes, all of whom reported the same result of 0.01609 ± 0.0008 for the total delayed neutron yield from thermal fission of ^{235}U. This agrees with the same result calculated in this work of 0.01609 ± 0.0008.

For the first set of results, additional information will be included that is not included in the other sections. Specifically, the delayed neutron count rate will be provided as well as a comparison to the count rate from calculated from other DNP group parameters in the literature. There will also be an example figure demonstrating how the stochastic sampling affects the DNP group parameters.

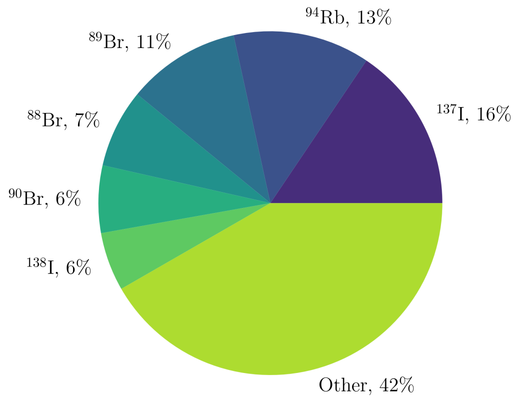


Fig. 1. The fraction of the delayed neutron yield from particular DNPs from JENDL-5 cumulative fission yields and IAEA beta-delayed neutron emission database half-lives and emission probabilities. The five DNPs with the largest delayed neutron yield contributions supply over 50% of the total delayed neutrons per fission event. The remaining 215 DNPs provide approximately 48% of the total delayed neutron yield.

III.A. IAEA Beta-Delayed Neutron Emission Database

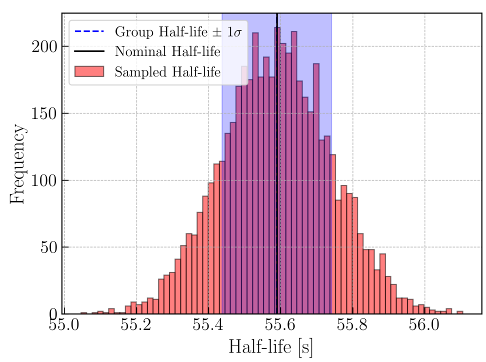
Figure 1 shows the percentage of the total delayed neutron yield that comes from the six DNPs with the largest yield contribution, where the yield from each DNP is defined by Equation (1). This equation shows that the largest yield contributors have large emission probabilities P_n and cumulative fission yields CFY. This doesn't necessarily indicate there will be a large concentration of the DNP, as it may be short lived, but rather for any given fission, that DNP has a higher probability to form and emit a delayed neutron. The total delayed neutron yield for this dataset is 0.0160 delayed neutrons per fission event.

Figure 2 shows the histogram for the sampled group one half-lives τ_1 as well as how the delayed neutron count rate compares to those using DNP group parameters from the literature [23, 6, 24]. The histogram shows that the group parameters follow a normal distribution as the nuclear data for each DNP is randomly sampled, with the group half-life as the mean of that normal distribution. Although not shown, the other group parameters follow this same trend. The Keepin normalized delayed neutron count rate shows how the delayed neutron count rate calculated in this work, both for every count rate from the sampled data and from the group fit using the calculated DNP group parameters compare. A value of 1 at any given time indicates the delayed neutron count rate at that time perfectly matches the delayed neutron count rate from the Keepin et al.

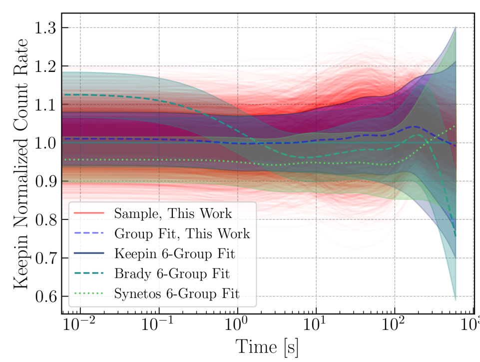
TABLE III
IAEA beta-delayed neutron emission database and JENDL-5 nuclear data delayed neutron yield and average half-life values.

Parameter	Value	Units
$\nu_d(I)$	0.0160 ± 0.0009	$[-]$
$\nu_d(K)$	0.0160 ± 0.0008	$[-]$
$\Delta\nu_d$	0.0263 ± 123	$[pcm]$
$\bar{\tau}(I)$	9.1 ± 0.5	$[s]$
$\bar{\tau}(K)$	9.0 ± 0.4	$[s]$
$\Delta\bar{\tau}$	0.01 ± 0.7	$[s]$

DNP group parameters. These results show that the Keepin et al. DNP group parameters agree well with group parameters calculated using the IAEA beta-delayed neutron emission database and JENDL-5 cumulative fission yields, with differences smaller than those from Brady and England and from Synetos and Williams.



(a) The DNP group one half-life τ_1 evaluated based on the mean of the sampled data.



(b) The delayed neutron count rate as a function of time relative to the Keepin et al. DNP group parameter delayed neutron count rate for DNP six group parameters from this work, Brady and England, and Synetos and Williams [23, 6, 24].

Fig. 2. These plots show how the 5000 samples lead to a distribution of DNP group parameters and delayed neutron count rates from JENDL-5 cumulative fission yields and IAEA beta-delayed neutron emission database half-lives and emission probabilities. The DNP group one half-life τ_1 shows a normal distribution about a mean, while the Keepin et al. normalized count rate shows this work offers good agreement with the literature.

Table III shows how the total delayed neutron yield ν_d and average half-life $\bar{\tau}$ compares between the value from the I DNPs and the six DNP groups. The differences between both are well within the uncertainty bounds, indicating the six groups are sufficiently capturing the general behavior of the delayed neutrons.

TABLE IV
IAEA beta-delayed neutron emission database and JENDL-5 nuclear data ten largest PCCs.

Nuclide	PCC_i
^{87}Br	3.678703
^{88}Br	3.397623
^{137}I	3.325644
^{96}Rb	3.277932
^{97}Rb	2.547497
^{94}Rb	2.157988
^{138}I	2.126428
^{89}Br	2.111075
^{90}Br	1.768819
^{91}Br	1.670366

Table IV shows the ten largest PCC_i terms, which reveals which DNPs have the greatest impact on the DNP group parameters $\nu_{d,k}$ and τ_k. This table shows that bromine alone has five isotopes that affect the DNP group parameters, with ^{87}Br being the dominant isotope due to its long-half life that makes it the primary DNP for the group one parameters. Generally, DNPs will have a large PCC_i for one of two reasons: they have a large delayed neutron yield and may affect several parameters moderately, such as ^{137}I; or they have a half-life that causes them to have a large impact on one specific group, such as ^{87}Br.

Table V shows the ten largest U_i terms, which corresponds to which DNPs have the largest uncertainties relative to their impact on the DNP group parameters $\nu_{d,k}$ and τ_k. These ten DNPs do not overlap at all with Table IV, indicating that the JENDL-5 cumulative fission yields and IAEA beta-delayed neutron emission database have small relative uncertainties for the DNPs in Table IV as otherwise those DNPs would dominate Table V. Interestingly, it appears that cobalt has several isotopes that have large U_i values, indicating that element has large relative uncertainties in its nuclear data relative to its impact on the DNP group parameters.

Figure 3 gives a more detailed view on which DNP values specifically lead to the large U_i values. This offers slightly more insight than Table V, which sums over the various DNP values. This figure shows that the half-lives τ and emission probabilities P_n are the primary drivers of U_i, at least for the ten most impactful DNPs. This indicates that the JENDL-5 cumulative fission yields offer sufficiently relative uncertainties compared to the relative uncertainties from the IAEA beta-delayed neutron emission database.

Figure 4 shows the PCC_i and U_i plotted on the chart of the nuclides. These plots do not limit

TABLE V
IAEA beta-delayed neutron emission database and JENDL-5 nuclear data ten largest uncertainty-weighted PCCs.

Nuclide	U_i
^{105}Zr	0.281379
^{72}Co	0.268531
^{116}Ru	0.260299
^{105}Y	0.246537
^{75}Co	0.244795
^{68}Co	0.235227
^{116}Rh	0.231495
^{71}Co	0.229786
^{99}Kr	0.226118
^{70}Co	0.222885

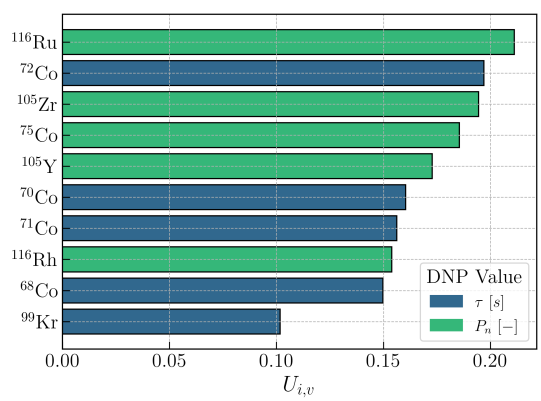
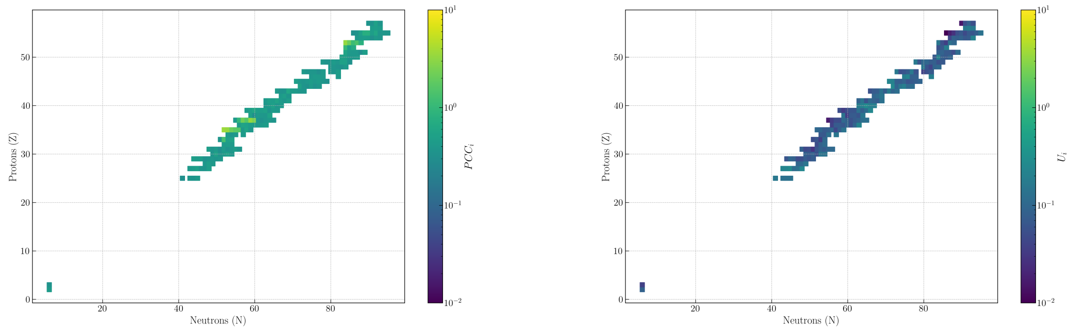


Fig. 3. The ten largest uncertainty-weighted PCCs $U_{i,v}$ sorted by DNP and associated DNP value from JENDL-5 cumulative fission yields and IAEA beta-delayed neutron emission database half-lives and emission probabilities.

to only the top ten largest values, but instead show how all of the values compare among each other. The values of U_i are all smaller than those of PCC_i, because all of the relative uncertainties are less than 1. There do not appear to be any clear trends, though in Figure 4(a), some horizontal bands of larger PCC_i values can be seen for 35 protons, 37 protons, and 53 protons; these correspond to bromine, rubidium, and iodine, respectively.



(a) The PCC_i, or total PCC, calculated for each DNP.

(b) The U_i, or total uncertainty-weighted PCC, for each DNP.

Fig. 4. Chart of nuclide plots showing the PCCs and uncertainty-weighted PCCs.

III.B. ENDF/B-VII.1

III.C. ENDF/B-VIII.0

III.D. JEFF-3.1.1

III.E. JENDL-5

III.F. Hybrid

IV. CONCLUSIONS

- Summarize the main takeaways from the results (maybe a table indicating DNPs of interest for different datasets?)
- Describe how improving these data will enable development of improved models of DNPs and kinetics models

Have a large table with all group parameters, yields, half-lives, etc. (maybe put full table in an appendix)

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